

Statistical Programming Languages

Exercises: Day 1

Introduction and Variables, Vectors, Matrices, etc.

Exercise 1 (My first R script):

Open a new document and save it as `Day_1_surname_name.R` (according to your own names) in a new folder called “Statistical Programming”. Answer the following questions in this file and indicate exercise numbers in comments.

- a) What is your working directory?
- b) Set your current working directory to “Statistical Programming”.
- c) Create two variables `str1` and `str2` containing the character strings “slumdog” and “millionaire”, respectively. Return them in the console. What’s their class? Are they vectors?
- d) Create a variable `x` with the value 34.9 and convert `x` into type character.

Save all your solutions for the following exercises in `Day_1_surname_name.R`. Use comments to structure your document.

Exercise 2 (R as a calculator):

Compute the following terms:

- a) $\sin(3\pi + 2)$
- b) $\log_5(10)$
- c) $\sum_{i=1}^{10} i$
- d) $\exp((2 + 3i)(4 - i))$
- e) $\sqrt[3]{482}$

Exercise 3 (Vectors):

- a) Create a vector `a` with the elements 1, 3, 5, 7, 11, 13, 17 and 19.
- b) Create a vector `b` with $2^1, 2^2, \dots, 2^8$.
- c) Create a vector `c` with $1^2, 2^2, \dots, 8^2$.
- d) Find the positions where elements of `b` and `c` coincide. (Hint: `?which`)

- e) Create a vector `d` which contains the numbers from 1 to 100 and another vector `e` which contains 100 elements equal to 7.
- f) Create a vector `f` with the first 30 elements of the following geometric progression: $a_n = 10 \cdot (1.5)^{n-1}$. Compute the cumulative sum of the vector elements. Give the elements and positions of `f` whose cumulative sum is greater than 10^6 .
- g) Compute the values of the function $y = e^{-x}$ for an equidistant grid from -3 to 3 with an stepwidth of 0.5.

Exercise 4 (Random vectors):

- a) Create a vector `norm` of size $n = 100$ from a normal distribution with mean 2 and standard deviation 4. Create two vectors called: `norm.even` and `norm.odd` with the elements of `norm` in the even and odd positions, respectively. Compute the mean and standard deviation of `norm.even` and `norm.odd`. Furthermore, compute the mean and standard deviation of `norm.even` and `norm.odd` considering only the positive numbers.
- b) Create a vector `pois` of size $n = 50$ from a poisson distribution with expectation 15. Create a new vector `pois3` that contains the elements in `pois` that are divisible by 3. Add random noise from a standard normal distribution to `pois3` and store it in a variable `pois3.noise`.

Exercise 5 (Matrices):

- a) Create a matrix `M.c` with the first column vector `b` and the second column vector `c` (from Exercise 3). Print the dimension and the seventh row of `M.c`.
- b) Create a matrix `M.r` with the first row vector `a` and the second row vector `b` (from Exercise 3). Rename the rows of `M.r` to “a” and “b” and the columns to “S”, “T”, ..., “Y”, “Z”. (Hint: These are the letters 19 to 26 of the alphabet. The alphabet is provided by the constants `letters` and `LETTERS`.)
- c) Print the matrix `M.r` without the column “W”.
- d) Print the elements of `M.r` larger than 12.
- e) Create a 10×10 matrix `D` which contains the numbers 1 to 100 filled by columns and another 10×10 matrix `E` which contains the numbers $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{100}$ filled by rows.
- f) Calculate the sum $D + E$, the difference $D - E$, the (matrix!) product $P = D \cdot E$ and the product of the elements $(d_{i,j} \cdot e_{i,j})_{i,j=1\dots 10}$. Print the diagonal elements of the matrix `P`.

Exercise 6 (Arrays):

The built-in R object `UCBAdmissions` is a 3-dimensional array with the number of admitted and rejected students at Berkeley University in 1973. The information is disaggregated by gender and department. Compute:

- a) the number of students admitted and rejected
- b) the number of males and females
- c) the number of students by departments
- d) the number of students admitted and rejected by gender
- e) the number of males and females by departments

(Advanced: Use the function `apply()` to solve this task, see `?apply`.)

Exercise 7 (Lists):

- a) Create a named list `List` with three components:

1. a vector `x` with 20 random numbers from a standard Normal distribution,
2. a 10×10 matrix `y` with 100 values from a standard Normal distribution,
3. a binary vector `z` with 10 zeros and 20 ones.

Compute the sum of each component. (Advanced: Use the function `lapply()` to compute the sums, see `?lapply`.)

- b) Create a vector `x` with 20 random values from a standard Normal distribution. Create a vector `y` applying the following linear function on `x`: $y = 3 + 2.5x$. Add to `y` a random noise through a centered Normal distribution with a standard deviation of 0.5. Fit a simple linear model using: `model <- lm(y~x)`. Verify that `model` is a list. Apply the `summary` function and access the list elements. Execute the following command `methods(class='lm')` to see all methods available for the class `lm`.